Yunsung Chung

PhD Candidate in Computer Science

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Professional Summary

Researcher in longitudinal EHR modeling, uncertainty-aware clinical AI, wearable biosignals, medical imaging, and clinically aligned generative models.

Education

PhD **Tulane University**, Computer Science — New Orleans, LA, USA
- Anticipated graduation: December 2026

Aug 2022 – Dec 2026

MS **The Australian National University**, Computer Science — Canberra, Australia
July 2019 – July 2021

BS **Dongguk University**, Electronics, Information Communication Engineering — Seoul, South Korea
Mar 2012 – Feb 2018

Experience

Samsung Research America, Research Intern — Mountain View, CA, USA
Designed algorithms for predicting clinical biomarkers from large-scale wearable PPG data under sparse and noisy labels.
May 2025 – Aug 2025
4 months

TRIAD Lab, Tulane University, Research Assistant — New Orleans, LA, USA
Lead projects on long-horizon EHR modeling, offline-RL and OPE benchmarking, synthetic medical data, cardiac imaging, and wearable signal modeling.
Aug 2022 – present
4 years 4 months

Airport Railroad Express, Hardware Engineer — Incheon, South Korea
Nov 2017 – May 2019
1 year 7 months

Publications

EHRDyn: A Benchmark for Offline Reinforcement Learning from EHRs with Complementary Simulators — 2027
Yunsung Chung, Han Feng, Nassir Marrouche, Jihun Hamm
(KDD)

Weighted Temporal Decay Loss for Learning Wearable PPG Data with Sparse Clinical Labels — 2026
Yunsung Chung, Keum San Chun, Migyeong Gwak, et al.
(ICASSP)

Auditing Long-Horizon Patient Trajectory Modeling in EHR Foundation Model Benchmarks — 2026
Yunsung Chung, Han Feng, Nassir Marrouche, Jihun Hamm
(Under review at NeurIPS)

CRAFT: Clinical Reward-Aligned Finetuning for Medical Image Synthesis — 2026
Yunsung Chung, Alex El Darzi, Carlo El Khoury, Han Feng, Nassir Marrouche, Jihun Hamm
(Under review at NeurIPS)

Intervention-Aware Clinical World Model for Post–Op Outcome Forecasting in Cardiology 2026
Yunsung Chung, Yingshuo Liu, Abboud F Hassan, Han Feng, Mary M Maleckar, Nassir Marrouche, Jihun Hamm
(MICCAI Workshop on Medical World Models (MWM) — Accepted)

SoK: Can Synthetic Images Replace Real Data? A Survey of Utility and Privacy of Synthetic Image Generation 2025
Yunsung Chung, Yunbei Zhang, Nassir Marrouche, Jihun Hamm
(USENIX Security)

SOFA: Deep Learning Framework for Simulating and Optimizing Atrial Fibrillation Ablation 2025
Yunsung Chung, Chanho Lim, et al.
(MICCAI)

Awards

FIRA RoboWorld Cup - 1st Place in Basketball 2016

International Robot Contest - 2nd Place in Marathon 2016

First-Class Academic Excellence Scholarship, Dongguk University 2012

Skills

Machine Learning
Advanced; PyTorch, MONAI, diffusion models, synthetic data, LoRA, preference optimization

Medical AI
Advanced; Cardiac MRI, ECG/PPG modeling, biomarker prediction, privacy-utility analysis

Service

Reviewer: ICASSP 2027; NeurIPS 2026; ICML 2026; MICCAI 2024–2026; IEEE TCSVT, 2023.

Languages

Korean
Native

English
Fluent